Telecom Customer Churn Analysis

Introduction

- This project focuses on Customer Churn Analysis in the Telecom Industry.
- Customer churn occurs when customers stop using a company's services. For telecom companies, churn directly affects revenue and customer growth.
- The objective of this project was to analyze customer data and identify patterns that lead to customer churn, so businesses can take actions to improve customer retention.

To complete this analysis, I used:

- **Python –** for data cleaning and preprocessing
  SQL – for querying and data analysis
  Power BI – for visualization and dashboard development

Dataset Collection

The dataset used in this project was obtained from **Kaggle**.
Dataset name: Telco Customer Churn Dataset
The dataset contains customer information from a telecom company and includes variables such as:

- Customer demographics
- Services used by customers
- Contract information
- Payment methods
- Monthly and total charges
- Customer churn status

The dataset contained **over 7000 customer records**, which provided sufficient data to analyze customer behavior patterns.

Understanding the Business Problem

- Before performing analysis, the first step was to understand the **business problem**.
- Telecommunication companies face a major challenge in retaining customers. When customers leave the service, the company loses revenue and must invest more resources to acquire new customers.

The business wanted to answer key questions such as:

- Why do customers leave the service?
- Which customers are more likely to churn?
- What factors influence churn behavior?
- How can the company reduce churn rates?

Understanding these factors helps companies develop **customer retention strategies**.

Defining the Business Objective

After understanding the business problem, the next step was defining the project objectives.

The main goals of the analysis were:

- Calculate the overall churn rate of customers.
- Identify key drivers that influence churn.
- Analyze customer behavior patterns.
- Segment customers based on demographics and services.
- Build an interactive dashboard that business users can easily understand.

Data Exploration

Once the dataset was collected, the next step was to explore the data.

During the exploration phase, the dataset structure was examined to understand:

- Number of rows and columns
- Types of variables
- Missing or inconsistent values
- Distribution of customer attributes

This helped identify areas where the data needed cleaning before performing analysis.

Data Cleaning and Preparation (Python)

- Before performing analysis, the dataset required cleaning and transformation.
- Python was used to perform these tasks efficiently.

The data cleaning process involved several steps.

Handling Missing Values

- During data inspection, it was observed that some columns contained missing or blank values.
- For example, the Total Charges column had empty entries for certain customers.
- These missing values were identified and handled so that the dataset could be used for accurate analysis.

Outcome:

- The dataset became complete and suitable for further analysis.

Data Type Correction

Some columns had incorrect data types.
For example:

- Numeric values stored as text

- Categorical values stored inconsistently
- These data types were corrected to ensure accurate analysis.

Outcome:

- All variables were converted into appropriate formats.

Feature Engineering

To make the analysis more meaningful, new features were created.
For example:

Tenure Groups

Customer tenure was grouped into categories such as:

0–1 Year
1–2 Years
2–4 Years
4–5 Years
5–6 Years

This allowed easier analysis of customer retention based on how long customers stayed with the company.

Total Services

Another feature was created to count the number of services each customer uses.
Customers may use multiple services such as:

- Internet
- Streaming TV
- Online Security
- Tech Support
- Device Protection
- Counting the number of services helped analyze whether customers with more services are less likely to churn.

Outcome:

- These new features improved analytical insights.

Data Analysis Using SQL

- After cleaning the dataset, SQL was used to perform structured data analysis.
- SQL queries were used to calculate key metrics and identify patterns.

Calculating Total Customers

- The total number of customers was calculated to understand the dataset size.
- 7043 total customers were present in the dataset.

Calculating Churn Rate

- The churn rate was calculated to determine the percentage of customers leaving the service.
- Approximately 26.49% of customers churned.
- This indicates that about one in four customers stopped using the service.

Contract Type Analysis

- Customers were grouped based on their contract types.
- Customers with month-to-month contracts showed the highest churn rates.
- This suggests that customers without long-term commitments are more likely to leave.

Internet Service Analysis

- Customers were analyzed based on internet service types.
- Fiber optic customers showed relatively higher churn rates.
- This indicates potential issues with pricing or service quality.

Data Visualization (Power BI)

- After completing the analysis, the cleaned dataset was imported into Power BI to build an interactive dashboard.
- The dashboard was designed to provide business stakeholders with clear insights.
- The dashboard consists of three pages.

Customer Churn Risk Story

This page provides a high-level overview of churn patterns.
Key metrics displayed include:

Customer Tenure vs Churn

- This visualization shows how churn varies across different tenure groups.
- Customers with shorter tenure have higher churn rates.
- This indicates that new customers are more likely to leave within the first few years.

Contract Type vs Churn

- This analysis compares churn rates across different contract types.
- Month-to-month contracts have the highest churn rates.
- Encouraging customers to switch to long-term contracts could reduce churn.

Internet Service vs Churn

- This visualization analyzes churn behavior across internet service types.
- Fiber optic users have relatively higher churn compared to other services.
- This suggests potential service or pricing concerns.

Customer Behavior Insights

This page focuses on customer behavior patterns.

Payment Method vs Churn

- This analysis examines churn patterns based on payment methods.
- Customers using electronic checks show higher churn.
- This may indicate lower customer commitment.

Monthly Charges vs Churn

- Customers were grouped based on monthly charges.
- Customers with higher monthly charges tend to churn more.
- Price sensitivity plays a role in churn behavior.

Number of Services vs Churn

- This analysis examines how service usage affects churn.
- Customers using fewer services are more likely to churn.
- Customers using multiple services are more engaged and likely to stay.

Customer Risk Segmentation

This page analyzes churn based on demographic factors.

Gender vs Churn

- Churn behavior is relatively similar across genders.
- Gender does not significantly influence churn.

Senior Citizen vs Churn

- Senior citizens show slightly different churn patterns compared to non-senior customers.

Partner vs Churn

- Customers without partners tend to churn more frequently.

Dependents vs Churn

- Customers without dependents have higher churn rates.
- Family status may influence service usage stability.

Key Business Insights

Tenure is the Strongest Retention Indicator: Customers in the 0–1 year tenure group show dramatically higher churn rates compared to long-term customers. This reveals a critical early engagement window if a customer survives the first year, their likelihood of staying increases significantly. The business is losing customers before they become profitable.

Contract Type Directly Controls Churn: Month-to-month contract customers churn at nearly 3x the rate of annual or two-year contract holders. These customers have no financial or commitment barrier to leaving, making them the highest-risk segment in the entire customer base.

Fiber Optic Customers Are Churning Despite Premium Pricing: Fiber optic users pay higher monthly charges yet show higher churn rates than DSL customers. This is a dangerous combination the company is losing its highest-revenue customers. This points to a service quality or value perception problem, not just a pricing issue.

Electronic Check Payment = Low Commitment Signal: Customers paying via electronic check churn significantly more than those using credit cards or bank transfers. Auto-pay customers tend to stay longer because payment becomes frictionless. Electronic check users are actively choosing to pay manually each time a behavioral signal of low commitment.

Price Sensitivity Among High-Charge Customers: Customers with monthly charges above $70 show noticeably higher churn rates. These customers are paying premium prices but may not feel they are receiving premium value creating a high churn-high revenue-loss risk for the business.

Service Bundling Protects Against Churn: Customers subscribed to 4 or more services churn far less than those with 1–2 services. Every additional service creates a "switching cost" making it harder and less convenient for customers to leave. Single-service customers have almost nothing keeping them tied to the provider.

Demographics Play a Secondary but Notable Role

- Senior citizens show slightly higher churn possibly due to pricing concerns or difficulty navigating digital services
- Customers without partners or dependents churn more frequently they have fewer household dependencies on the service and switch more easily
- Gender has no significant impact on churn gender-based targeting would be a wasted investment

Business Recommendations

Launch an Aggressive Early Retention Program (0–12 Months): The data clearly shows the first year is the most critical period. The business should:

- Assign a dedicated onboarding journey for new customers with proactive check-ins at 30, 60, and 90 days
- Offer loyalty rewards or bill credits after the first 3 months to reinforce commitment
- Use automated churn prediction alerts to flag at-risk new customers before they decide to leave

Expected Impact: Reducing early churn by even 10–15% could significantly improve customer lifetime value across the base.

Create Strong Incentives to Convert Month-to-Month Customers: Month-to-month customers represent the single largest churn risk pool. The business should:

- Offer meaningful discounts (15–20%) for customers who switch to annual or two-year contracts
- Run targeted campaigns specifically toward month-to-month customers who have been with the company 3–9 months before they hit peak churn risk
- Introduce a "Contract Lock-In Bonus" extra data, free streaming service, or discounted add-ons for signing a longer contract

Expected Impact: Even a 20% conversion of month-to-month customers to annual plans could substantially reduce overall churn rate.

Investigate and Fix the Fiber Optic Experience: Fiber optic customers are the company's most valuable segment and they are leaving. The business must:

- Conduct customer satisfaction surveys specifically for fiber optic users to identify pain points (speed issues, outages, support quality)
- Benchmark fiber optic pricing against competitors if the price is not justified by perceived quality, customers will switch
- Create a "Fiber Premium" loyalty tier with priority support, guaranteed speed SLAs, and exclusive benefits to reduce churn in this segment
- Review whether marketing promises match actual service delivery

Expected Impact: Retaining even a fraction of high-paying fiber optic churners directly protects the company's highest revenue stream.

4. Drive Auto-Pay Adoption to Reduce Churn Passively: Since electronic check users churn more, converting them to automatic payment is a low-cost, high-impact strategy:

- Offer a small monthly bill discount (e.g., $5–$10 off) for enrolling in auto-pay via credit card or bank transfer
- Send targeted communications to electronic check users highlighting the convenience and savings of switching
- Make auto-pay enrollment a default option during sign-up for new customers

Expected Impact: Auto-pay customers churn less because payment friction is removed, and the discount creates perceived value. This is one of the highest ROI retention tactics available.

Implement a Service Bundling Strategy: Since multi-service customers churn far less, the business should actively push bundling:

- Introduce "Starter Bundle" packages that combine internet + phone + one additional service at a discounted rate
- For customers using only 1–2 services, send personalized upgrade offers showing cost savings of adding more services
- Train customer service teams to proactively upsell relevant services during every customer interaction
- Create a loyalty milestone program customers who maintain 4+ services for 12 months receive a meaningful reward

Expected Impact: Each additional service a customer adopts reduces their churn probability and increases monthly revenue simultaneously.

Develop a Targeted Senior Citizen Retention Program: Senior citizens show higher churn and are often underserved by standard digital-first support:

- Create a dedicated senior support line with extended hours and patient, non-technical agents
- Offer simplified billing and plan options designed specifically for this segment
- Provide in-home or in-store tech assistance programs to help senior customers get more value from their services

Expected Impact: Improved satisfaction for a segment that, once loyal, tends to stay for very long periods.

Build a Real-Time Churn Prediction System: The data from this project provides enough signal to build a proactive churn prediction model that:

- Scores every customer monthly based on tenure, contract type, payment method, number of services, and monthly charges
- Automatically triggers retention interventions (personalized offers, callbacks, loyalty rewards) for customers flagged as high-risk
- Allows the retention team to prioritize outreach toward high-value customers who are showing early churn signals

Expected Impact: Moving from reactive (retaining customers after they call to cancel) to proactive (engaging at-risk customers before they decide to leave) is the single most powerful long-term churn reduction strategy.

Overall Conclusion:

This analysis reveals that churn is not random it is predictable and preventable. The customers most likely to churn share identifiable characteristics: new, on short-term contracts, paying high prices, using few services, and paying manually. By targeting these specific segments with the right interventions at the right time, the business can materially reduce churn, protect revenue, and build a more loyal customer base.